2020/2021

2020 / 2021.

Q1) a) i, $764A_{16}$

Binary

7 : 6 : 4 : A

0111 : 0110 : 0100 : 1010

0111011001001010_2

Octal

0|111|011|001|001|010

0 7 3 1 1 2

73112_8

ii $89 45FA6B_{16}$

Binary.

$1000\ 1001\ 0100\ 0101\ 1111\ 1010\ 0110\ 1011_2$

Octal

10|000 1|001|0 10|0 01|01 1|111|101|0 01|10 1|011

2 1 1 2 1 3 7 5 1 5 3

21121375153_8

iii, $18AF67B_{16}$

Binary

$0001\ 1000\ 1010\ 1111\ 0110\ 0111\ 1011_2$

Octal

0|001|100|0 10|10 1|111|011|0 01|111|011

0 1 4 2 5 7 3 1 7 3

142573173_8

b) i, 11010101_2 = 213_{10}

ii, 11010.110_2 = ~~26.0001~~$_2$

$2^4 2^3 2^2 2^1 2^0$ $2^{-1} 2^{-2} 2^{-3}$

16 + 8 + 2 0.5 + 0.25

26 = 0.75

26.75_{10}

~~0.110 × 2~~

~~0.220 × 2~~

~~0.440 × 2~~

~~0.880 × 2~~

~~1.760~~

~~0.760 × 2~~

~~1.520~~

iii, $0100\ 11110.1010_2$ = 158.5

c) 67.567

1000011.10010$_2$

$$\begin{array}{r|l} 2 & 67 \\ 2 & 33-1 \\ 2 & 16-1 \\ 2 & 8-0 \\ 2 & 4-0 \\ 2 & 2-0 \\ & 1-0 \end{array}$$

0.567 × 2 = 1.134 → 1

0.134 × 2 = 0.268 → 0

0.268 × 2 = 0.536 → 0

0.536 × 2 = 1.062 → 1

0.062 × 2 = 0.124 → 0

~~d) i, -16 + 78~~

~~16/ 10000~~

~~00010000~~

~~11101111~~

~~+1~~

~~11110000~~

$$\begin{array}{r|l} 2 & 16 \\ 2 & 8-0 \\ 2 & 4-0 \\ 2 & 2-0 \\ & 1-0 \end{array} \qquad \begin{array}{r|l} 2 & 78 \\ 2 & 39-0 \\ 2 & 19-1 \\ 2 & 9-1 \\ 2 & 4-1 \\ 2 & 2-0 \\ & 1-0 \end{array}$$

~~11110000~~

~~+ 1001110~~

~~Sign bit → 10011110~~

~~convert into 2s' c. 10011110~~

~~011000001~~

~~+1~~

~~011000010$_2$~~

d) i, -16 + 78

16/ 00010000$_2$ ⇒ 1s complement 11101111$_2$

Addition.

11101111$_2$

\+ 01001110$_2$

100111101$_2$

100111101$_2$

011000010$_2$

ii, -90 - 8

90/ 01011010$_2$ ⇒ 10100101$_2$

8/ 1000$_2$

$$\begin{array}{r|l} 2 & 90 \\ 2 & 45-0 \\ 2 & 22-1 \\ 2 & 11-0 \\ 2 & 5-1 \\ 2 & 2-1 \\ & 1-0 \end{array} \qquad \begin{array}{r|l} 2 & 8 \\ 2 & 4-0 \\ 2 & 2-0 \\ & 1-0 \end{array}$$

~~10100101~~

~~+ 11110111~~

~~110011100$_2$~~

10100101

\- 1000

10011101

10011101 ⇒ 1s compl.

01100010$_2$

= -98$_{10}$

2 | 13	2 | 26
2 | 6 − 1	2 | 13 − 0
2 | 3 − 0	2 | 6 − 1
1 − 1	2 | 3 − 0
	1 − 1

d)

iii) 13 − 26

~~00001101~~

~~− 00011010~~

~~10011~~

00011010_2

1s' → 11100101_2

+ 1101_2

11110010

$00001101_2 = -13_{10}$

2 | 23	2 | 7	2 | 15	2 | 10
2 | 11 − 1	2 | 3 − 1	2 | 7 − 1	2 | 5 − 0
2 | 5 − 1	1 − 1	2 | 3 − 1	2 | 2 − 1
2 | 2 − 1		1 − 1	1 − 0
1 − 0			

e) i) 15 − 7

$7 / 00000111_2 \Rightarrow$

~~11111000_2~~
~~$+1_2$~~
~~11111001_2~~
~~$+ \quad 111_2$~~
~~100000000~~
~~011111111~~
~~$+1$~~
~~100000000~~

00001111
-00000111
$00001000_2 = 8_{10}$

ii) −10 + 6

00001010_2
11110101
$+1$
11110110

11110110
$+ \quad 0110$
11111100
00000011
$+1$
$00000100_2 = -4_{10}$

iii) −23 − 7.

00010111
11101000
$+1$
11101001_2

~~00000111~~
~~11111000~~
~~$+1$~~
~~11111001_2~~

~~11101001~~
~~11111001~~

11101001_2
$+ \quad 111$
11110000
00001111
$+1$
$00010000_2 = -16_{10}$

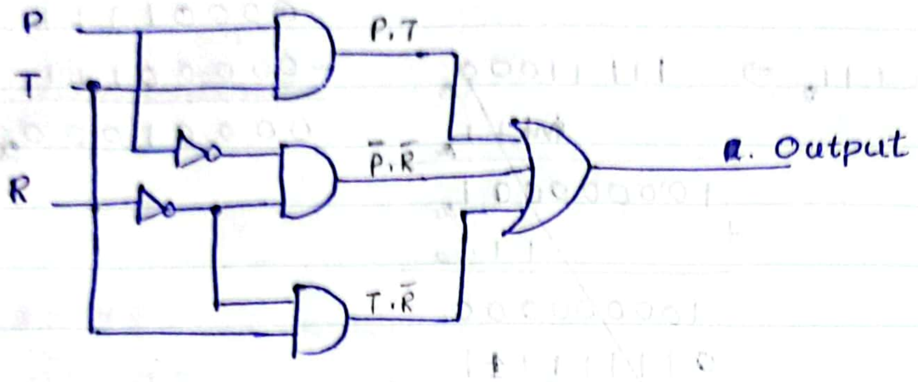
P T R $(P.T)+(\bar{P}.\bar{R})+(T.\bar{R})$

1 and 1

0 and 0

1 0

Q2) a) i)



ii) Truth table.

P	T	R	(P.T)	$(\bar{P}.\bar{R})$	$(T.\bar{R})$	Output
0	0	0	0	1	0	1
0	0	1	0	0	0	0
0	1	0	0	1	1	1
0	1	1	0	0	0	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	1	0	1	1
1	1	1	1	0	0	1

iii) Output $= (\bar{P}.\bar{T}.\bar{R}) + (\bar{P}.T.\bar{R}) + (P.T.\bar{R}) + (P.T.R)$

iv) $(\bar{P}\bar{T}\bar{R}) + (\bar{P}T\bar{R}) + (P.T.\bar{R}) + (P.T.R)$

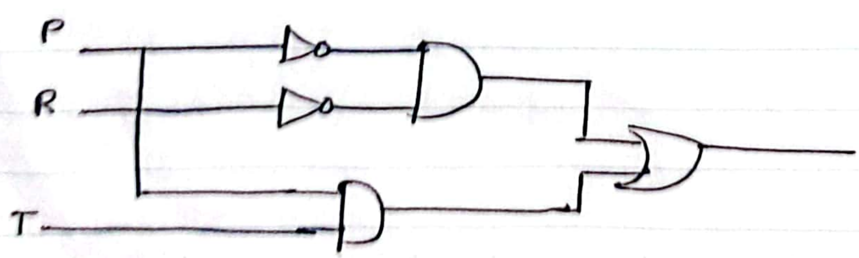
$\cancel{\bar{R}(\bar{P}\bar{T} + \bar{P}T + PT)}$

$= \bar{P}\bar{R}(\cancel{\bar{R}} + \bar{T} + T) + PT(\bar{R}+R)$

$= \bar{P}\bar{R}\ (1) + PT\ (1)$ $[\bar{A}+A=1]$

$= \bar{P}\bar{R} + PT$

v)



b) i) $F = (A+B)(A+\bar{B})(\bar{A}+C)$

$= A.((A+\bar{B})(\bar{A}+C)) + B.(A+\bar{B})(\bar{A}+C)$ (Distributive)

$= (A.A + A.\bar{B})(\bar{A}+C) + (A.B + B\bar{B})(\bar{A}+C)$ (Distributive)

$= (A + A.\bar{B})(\bar{A}+C) + (A.B + 0)(\bar{A}+C)$ (Distributive)

$= A\bar{A} + AC + A\bar{A}\bar{B} + A\bar{B}C + A\bar{A}B + ABC$ $[A.\bar{A} = 0]$

$= 0 + AC + 0 + A\bar{B}C + 0 + ABC$ $[A+\bar{A} = 1]$

$= AC(1 + \bar{B} + B)$ $[A+\bar{A} = 1]$

$= AC(\cancel{2})$ $[A.1 = A]$

$= AC$

ii) $F = \bar{A}B + B\bar{C} + BC + A\bar{B}\bar{C}$

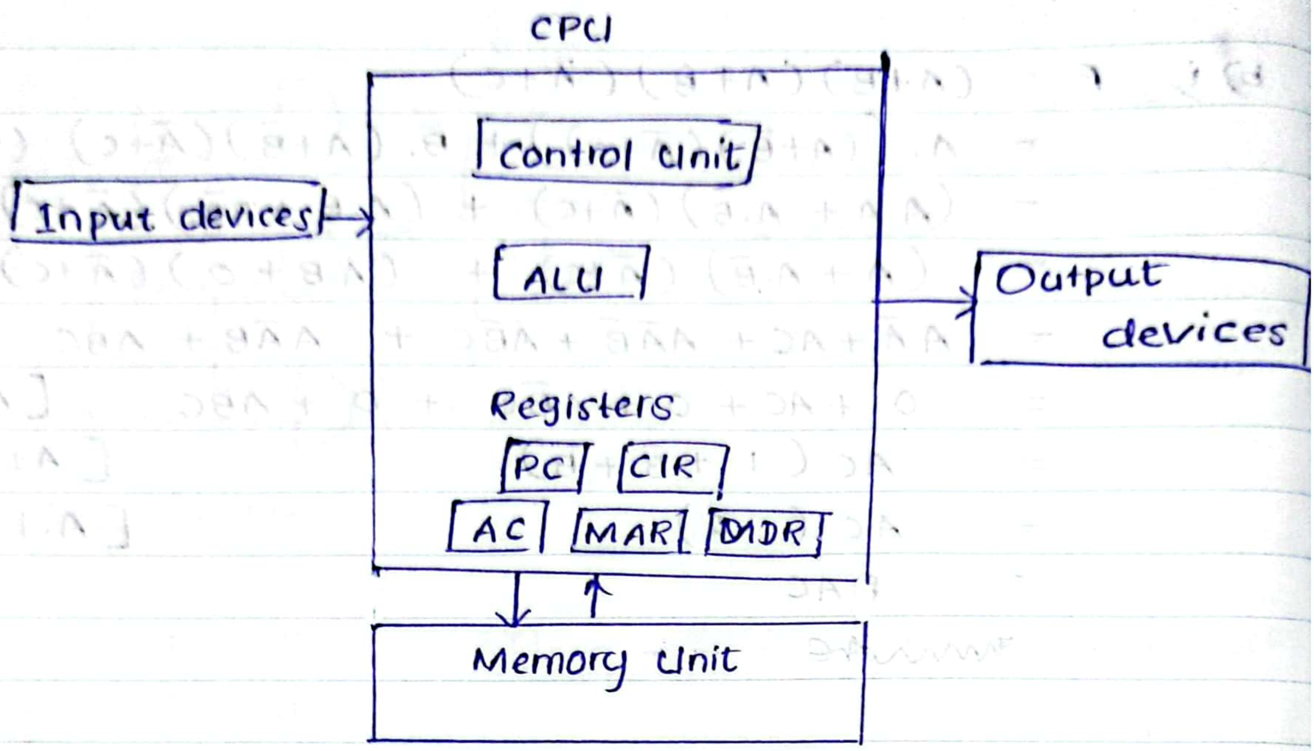
$= B(\bar{A} + \bar{C} + C) + A\bar{B}\bar{C}$ (Distributive law)

$= B(\bar{A} + 1) + A\bar{B}\bar{C}$ $(A + \bar{A} = 1)$

$= B(1) + A\bar{B}\bar{C}$ $(A + 1 = 1)$

$= B + A\bar{B}\bar{C}$

Q3) a)



d) Storing data.
They hold ~~the~~ important information that the CPU needs to work with, like numbers of instructions. they can hold data temperorily while CPU ~~is~~ is busy doing calculations or processing information.

~~Moving data around. :~~
Fast Access storage.
They are built directly into the chip itself, So the CPU can access them super quickly. This locations store important data and instructions that the CPU needs to work with right away.

Immediate data for ALU
These memory locations hold data that the ALU needs immediately for its calculations, so when the CPU is adding, subtracting, ... or do any other math, it can grab the numbers it needs from these memory locations right away without waiting for data from elsewhere. This helps the CPU work fast.

e) i) cashe memory.

It's a high speed memory sits between CPU and RAM. It'll temperorily store frequently accessed data and instructions. It'll reduce the time. It act as a buffer between CPU and RAM. and reduce the total execution time of the program.

ii) Main memory.

Primary memory or system memory. Information is accessed randomly instead of sequentially. Access times are much faster. It's a volatile memory. If the computer is turned off, all data contained in RAM is lost. There are several types of RAMs.

Ex:- EDO RAM, DDR, DDR 3

SD RAM, DDR 2, DDR 4

iii) secondary memory.

It serves as long-term storage for data. It provides non-volatile storage for documents, programs and multimedia files.

slower than the primary memory. Expanding the computer's storage capacity.

Ex:- HDD, SSD, optical discs, magnetic tapes, USB flash drives.

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Q4)

a) Flynn proposed a classification for computer architecture based on the number of instruction streams and data streams. Flynn uses stream concept for describe. a machine structure. A strea means,

Instruction stream

data stream.

b)

CISC	RISC
Memory to memory operations	Register to Register operations.
Emphasis on hardware	Emphasis on software.
small code sizes and high cycle per second	Large code size and lowe cycles per second.

c) It store the memory address of the next instructio to be executed. As the CPU executes each instruction the PC automatically moves to the next instruction memory address. This ensures that instructions are executed in the correct sequence and allows for branching to different parts of the program when needed. The PC guides the CPU through the program's execution process.

d) Shared memory means that multiple parts of a computer system can access the same memory at the same time. This allows different processes or threads to share data and communicat~~ore~~ with each other easily.

e) A technique used in advanced microprocessors where the microprocessor begins executing a second instruction before the first has been completed.

f) i) Immediate addressing mode.

The operand is directly specified in the instruction itself. You directly tell the CPU what the data is.

ex :- ADD 5, where 5 is the data

ii) Direct addressing mode.

You tell the CPU where to find the data.

ex :- LOAD from memory address 200. The address is directly specified in the instruction.

iii) indirect addressing mode.

You tell the CPU where to find the address of the data.

Ex :- LOAD From the memory address stored at memory address 300

iv) indexed addressing mode.

You tell the CPU to find the data by adding an index to a base address.

ex :- LOAD from memory address x plus 5, where x is a base address stored in a register

2019/2020

No:............ Date:....../....../......

2019 /2020

Q1) a)

i) $87AD_{16}$

I. $1000\ 0111\ 1010\ 1101_2$

II. 103655_8

ii) $9E7A2B_{16}$

I. $1001\ 1110\ 0111\ 1010\ 0010\ 1011_2$

II. 47475053_8

iii) $B64FDE_{16}$

I. $1011\ 0110\ 0100\ 1111\ 1101\ 1110_2$

II. 55447736_8

b) i) $1001101_2 = 77_{10}$

ii) $1010011.101_2 = 83 + 0.625 = 83.625_{10}$

iii) $10101110.1001_2 = 174 + 0.5625 = 174.5625_{10}$

c) 43.167

$$\begin{array}{l|l} 2 & 43 \\ 2 & 21-1 \\ 2 & 10-1 \\ 2 & 5-0 \\ 2 & 2-1 \\ & 1-0 \end{array}$$

0.167 × 2 = 0.334 → 0

0.334 × 2 = 0.668 → 0

0.668 × 2 = 1.336 → 1

0.336 × 2 = 0.672 → 0

0.672 × 2 = 1.344 → 1

101011.00101_2

d) i) $123_{10} = 01111011_2$

$$\begin{array}{l|l} 2 & 123 \\ 2 & 61-1 \\ 2 & 30-1 \\ 2 & 15-0 \\ 2 & 7-1 \\ 2 & 3-1 \\ & 1-1 \end{array}$$

ii) $-96_{10} = 01100000_2$

+1

~~01100001_2~~

10011111_2

+1

10100000_2

$$\begin{array}{l|l} 2 & 96 \\ 2 & 48-0 \\ 2 & 24-0 \\ 2 & 12-0 \\ 2 & 6-0 \\ 2 & 3-0 \\ & 1-1 \end{array}$$

2 | 8 .
2 | 4 - 0
2 | 2 - 0
1 - 0

2 | 5
2 | 2 - 1
1 - 0

2 | 4
2 | 2 - 0
1 - 0

2 | 14
2 | 7 - 0
2 | 3 - 1
1 - 1

2 | 13
2 | 6 - 1
2 | 3 - 0
1 - 1

e) i) $5 + 4$

$$\begin{array}{r} 00101_2 \\ +\ 00100_2 \\ \hline 01001_2 \end{array}$$

ii) $-14 + 5$

$$\begin{array}{r} 01110 \\ 10001 \\ +1 \\ \hline 10010_2 \end{array} \qquad \begin{array}{r} 10010 \\ +\ 00101 \\ \hline 10111_2 \end{array}$$

iii) $13 - 7$

$$\begin{array}{r} \cancel{00111} \\ \cancel{11000} \\ \cancel{+1} \\ \hline \cancel{11001_2} \end{array} \qquad \begin{array}{r} 01101_2 \\ -\ 00111_2 \\ \hline 00110_2 \end{array}$$

iv) $-8 - 4$

$$\begin{array}{r} 01000 \\ 10111 \\ +1 \\ \hline 11000_2 \end{array} \qquad \begin{array}{r} 11000_2 \\ -\ 00100_2 \\ \hline 01100_2 \end{array}$$

Q2) a) $F = \bar{A}\bar{B}\bar{C} + ABC + A\bar{B}\bar{C} + \bar{A}BC + ABC$

$= \bar{B}\bar{C}(\bar{A} + A) + BC(A + \bar{A} + A)$ [Distributive]

$= \bar{B}\bar{C}(1) + BC(1)$ [$A + \bar{A} = 1$] [$A \cdot 1 = A$

$= \bar{B}\bar{C} + BC$

b) i)

ii) $F_0 = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + ABC$

$F_1 = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$

$F_2 = \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC.$

iii) $F_0 = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + ABC$ [Distributive]

$= \bar{A}\bar{C}(\bar{B}+B) + AB(\bar{C}+C) + A\bar{B}\bar{C}$ [$\bar{A}+A=1$]

$= \bar{A}\bar{C}(1) + AB(1) + A\bar{B}\bar{C}$ [$A.1=A$]

$= \bar{A}\bar{C} + AB + A\bar{B}\bar{C}$

$F_1 = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$ [Distributive]

$= \bar{A}\bar{B}(\bar{C}+C) + BC(\bar{A}+A) + A\bar{B}C$ [$\bar{A}+A=1$]

$= \bar{A}\bar{B}(1) + BC(1) + A\bar{B}C$ [$A.1=A$]

$= \bar{A}\bar{B} + BC + A\bar{B}C$

$= \bar{A}\bar{B}$

$F_2 = \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC.$ [Distributive]

$= \bar{A}\bar{B}C + AC(\bar{B}+B) + A\bar{C}(\bar{B}+B)$ [$\bar{A}+A=1$]

$= \bar{A}\bar{B}C + AC(1) + A\bar{C}(1)$ [$A.1=A$]

$= \bar{A}\bar{B}C + AC + A\bar{C}$ [Distributive]

$= \bar{A}\bar{B}C + A(C+\bar{C})$ [$\bar{A}+A=1$, A.

$= \bar{A}\bar{B}C + A$

iv)

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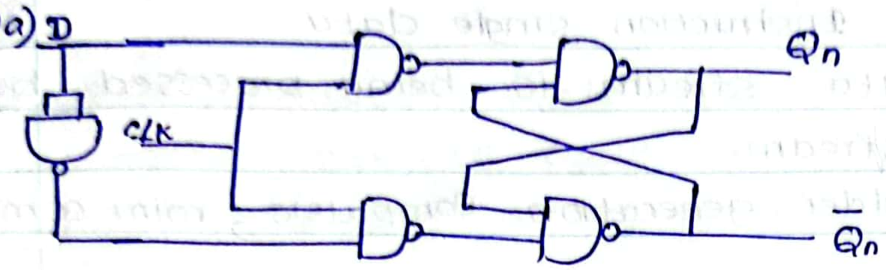
Q3) a)



Truth table.

CLK	D	Q_n	$\bar{Q}_n$
0	X	No change	
1	0	0	1
1	1	1	0

b) Registers

c) i) Memory Address Register.

It is a special type of Register. in a computer's CPU. Its main function is to hold the memory address of the data that the CPU wants to access in the computer's memory. (It holds the address of the data it wants to find in the computer's memory)

ii) PC

iii) Memory Data Register

It is like a temporary storage box in a computer's brain. It holds the actual data that the computer is reading from or writing to the memory. It acts as a bridge between the CPU and the memory.

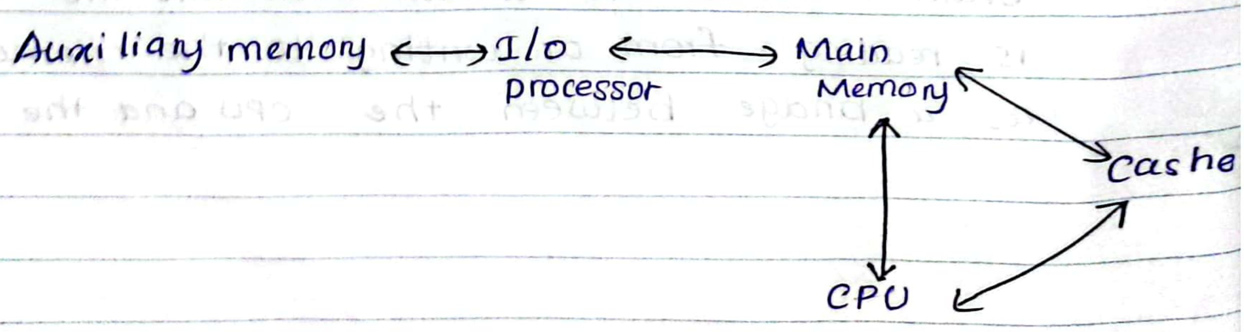
d) SISD : single Instruction single data.
A single data stream is being processed by one instruction stream.
ex: Older generation Computers, mini compu--

SIMD : single instruction Multiple data
Each instruction is executed on a different set of data by different processors under the supervision of a common control unit.

MISD : Multiple instruction single data.
each processor executes a different sequence of instructions.
Non existent.

MIMD : Multiple instructions multiple data.
It have multiple processing units, each with its own instruction stream and data stream.

e) Main memory directly communicates with CPU. and with auxiliary memory throug I/O processors.
Cashe is used to increase the speed of processing by making current programs and data available to the CPU at a rapid rate



Q4)
a)

CISC	RISC
Memory to memory operations	Register to Register operations.
Emphasis on hardware	Emphasis on software.
small code sizes and high cycle per second	Large code size and lowest cycles per second.

b) ~~Also~~ Instruction cycle is also known as fetch - decode - execute cycle. It is the fundamental process by which a microprocessor executes instructions. It involves 3 main steps.

Fetch: processor fetches the next instruction from memory.

Ex: let's consider an instruction "ADD R1, R2, where R1, R2 are registers.

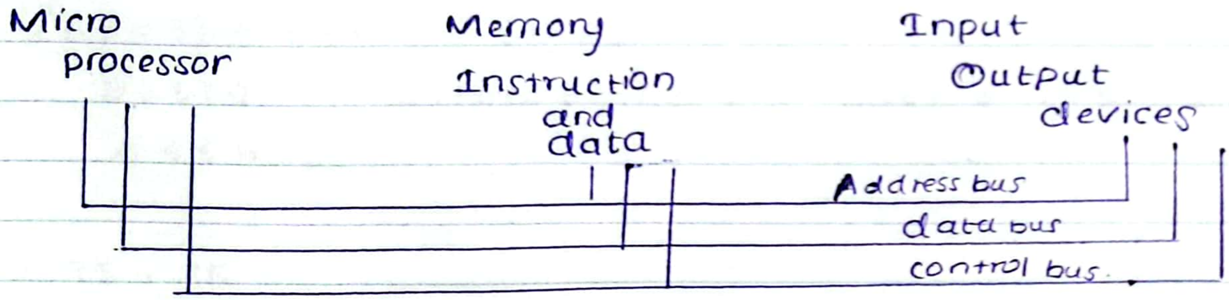
Decode: It interprets the instruction to determine what operation needs to be performed and what data it needs. ex:- (ADD, LOAD, etc...)

"ADD R_1, R_2"

execute: The processor carries out the instruction, performing the specific operation on the data

Eg:- Add R1, R2 and store the results back into reg. R1

c)



Address bus is a set of wires carries memory addresses from the CPU to ~~other~~ Main memory & I/O devices.

Control bus is a set of wires that carries control signals between different components of the CPU

Data bus is a set of wires that carries the data between CPU and other components d)

Immediate addressing mode.

The operand is directly specified in the instruction itself. You directly tell the CPU what the data is.

ex :- ADD 5 , where 5 is the data

Direct addressing mode.

You tell the CPU where to find the data.

ex :- LOAD from memory address 200. The address is directly specified in the instruction.

indirect addressing mode.

You tell the CPU where to find the address of the data.

Ex :- LOAD from the memory address stored at memory address 300

e) i) Virtual Memory.

It is a method of using computer harddrive to provide extra memory for the computer. When the computer runs out of physical memory (RAM) It uses some space ~~s~~ on the hard drive to store less frequen[illegible] used data temporarily.

ii) shared memory.

Shared memory means that multiple parts of a
computer system can access the same memory
at the same time. This allows different processes
or threads to share data and communicat~~tor~~e with
each other easily.

iii) Distributed memory.

In distributed memory systems, each processor has its own private memory. Communication between processors happen through message passing.

2018/2019

2018 - 2019.

$$\begin{array}{r|l} 2 & 95 \\ 2 & 47-1 \\ 2 & 23-1 \\ 2 & 11-1 \\ 2 & 5-1 \\ 2 & 2-1 \\ & 1-0 \end{array} \qquad \begin{array}{r|l} 2 & 15 \\ 2 & 7-1 \\ 2 & 3-1 \\ & 1-1 \end{array}$$

Q1) a) i) A C 5 B 4

$1010\,1100\,0101\,1011\,0100_2$

2542664_8

2 5 4 2 6 6 4

$8^6\ 8^5\ 8^4\ 8^3\ 8^2\ 8^1\ 8^0$

$= (262144\times2)+(32768\times5)+(4096\times4)+(512\times2)+(64\times6)+(8\times6)+(1\times4)$

$= 524288 + 163840 + 16384 + 1024 + 384 + 48 + 4$

$= 705972.$

ii) 6 C 8 D A

$0110\,1100\,1000\,1101\,1010_2$

1544332_8

1 5 4 4 3 3 2

$8^6\ 8^5\ 8^4\ 8^3\ 8^2\ 8^1\ 8^0$

$= (262144\times1)+(32768\times5)+(4096\times4)+(512\times4)+(64\times3)+(8\times3)+(1\times2)$

$= 262144 + 163840 + 16384 + 2048 + 192 + 24 + 2$

$= 444634.$

b) i) $-95 + 15$

$$\begin{array}{r} 01011111 \\ 10100000 \\ +1 \\ \hline 10100001 \end{array} \qquad \begin{array}{r} 10100001 \\ +\quad 1111 \\ \hline 10110000 \\ 01001111 \\ +1 \\ \hline 01010000_2 \end{array}$$

$$\begin{array}{r|l} 2 & 22 \\ 2 & 11-0 \\ 2 & 5-1 \\ 2 & 2-1 \\ & 1-0 \end{array} \quad \begin{array}{r|l} 2 & 82 \\ 2 & 41-0 \\ 2 & 20-1 \\ 2 & 10-0 \\ 2 & 5-0 \\ 2 & 2-1 \\ & 1-0 \end{array} \quad \begin{array}{r|l} 2 & 46 \\ 2 & 23-0 \\ 2 & 11-1 \\ 2 & 5-1 \\ 2 & 2-1 \\ & 1-0 \end{array} \quad \begin{array}{r|l} 2 & 122 \\ 2 & 61-0 \\ 2 & 30-1 \\ 2 & 15-0 \\ 2 & 7-1 \\ 2 & 3-1 \\ & 1-1 \end{array}$$

ii) $-82 + 46$

01010010
10101101
+1
10101110_2
+ 101110
11011100_2
00100011
+1
00100100_2

iii) $-122 - 22$

01111010
10000101
+1
10000110
− 10110
01110000
10001111
+1
10010000_2

c) $789 = 0111\ 1000\ 1001_2$

$12483 = 0001\ 0010\ 0100\ 1000.0011_2$

d) It has a limited character set (128 chars)
Doesn't encode many symbols.
Doesn't coded for other languages.

e) i) $0.1011_2 = 0 + 0.5 + 0.1875$
$= 0.6875_{10}$

ii) $1111.01001 = 15 + 0.28125$
$= 15.28125_{10}$

Q2)

a) A combinational circuit is a digital circuit that gives an output based only on its current inputs, That gives you an answer right away without remembering past operations.

Characteristics

→ input - dependent output - Output solely determined by its inputs.

→ No memory - It doesn't remember past input or outputs, each output depend on curr. Inpu

→ Logic gats - It's build using logic gates

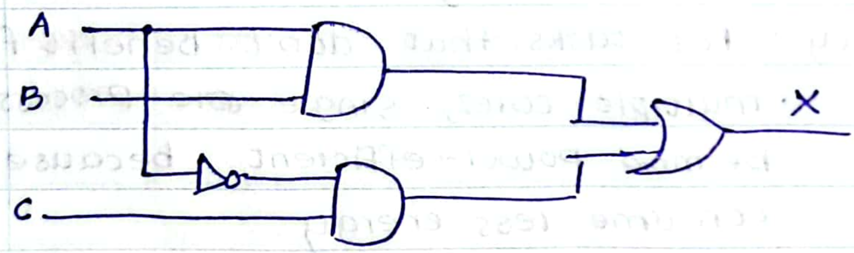
→ No feedback loops - There are no loops in the circuit when the output feeds back into the input.

→ Fast operation - Since it doesn't have memory, It can process inputs quictly.

b)

$$X = (\bar{A}.B).(\bar{A}.\bar{C}) + (B.C)$$

c)



d) i)

A	B	C	G_1	G_2	G_3	G_4	F G_5	G_6	G_7	G_8	G
0	0	0	1	0	0	0	1	0	0	1	1
0	0	1	1	0	0	0	1	0	0	1	0
0	1	0	1	0	1	0	1	0	0	1	1
0	1	1	1	1	1	0	1	1	1	1	0
1	0	0	0	0	1	0	1	0	0	1	1
1	0	1	0	0	1	0	1	1	1	1	0
1	1	0	0	0	1	1	0	0	0	0	1
1	1	1	0	0	1	0	1	1	1	1	0

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d) ii, $\cancel{\overline{(A\cdot B\cdot \bar{C})} + ((A+B)\cdot C)\cdot C + (\bar{A}\cdot B\cdot C)}$

$$F = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + ABC$$

$$= \bar{A}\bar{B}(\bar{C}+C) + BC(\bar{A}+A) + A\bar{B}(\bar{C}+C) + \bar{A}B\bar{C}$$

$$= \bar{A}\bar{B}(1) + BC(1) + A\bar{B}(1) + \bar{A}B\bar{C}$$

$$= \bar{A}\bar{B} + BC + A\bar{B} + \bar{A}B\bar{C}$$

$$= \bar{B}(\bar{A}+A) + BC + \bar{A}B\bar{C}$$

$$= \bar{B} + BC + \bar{A}B\bar{C}$$

Q3) a)

LOAD 658	LOAD 1200	LOAD 789
MULT 690	DIV 101	ADD 1111
STORE 789	STORE 1111	STORE 65

b) the little endian system, is a method of storing multiple data types in computer memory.

c)

CISC	RISC
Memory to memory operations	Register to Register operations.
Emphasis on hardware	Emphasis on software.
small code sizes and high cycle per second	Large code size and lowe cycles per second.

d) simplicity : Single-core processors are easier to design and program because there's only one core to manage.

compatibility : Some older software works better on single-core processors since it isn't optimized for multicore systems.

Power efficiency : For tasks that don't benefit from multiple cores, single-core processors can be more power-efficient because they consume less energy

specialized Applications : In certain specialized applications where parallel processing isn't necessary or practical, such as embedded systems or simple electronic devices, s-c-p might be more than sufficient to handle the workload e)

Instruction Fetch	→ Obtain instruction from program storage
↓	
Instruction decode	→ Determine required actions and instruction size.
↓	
Operand fetch	→ Locate and obtain operand data
↓	
Execute	→ Compute result value and status
↓	
Writeback results.	→ Deposit results in storage for later use

Q4)

SSD	HDD
store data using flash memory chips, with no moving parts	store data on magnetic disks than spin at high speed.
Faster than HDD. due to their lack of mechanical components	slower than SSD. due to the mechanical nature of their operation
durable. because they're not susceptible to damage from physical shock or vibration.	more prone to damage from physical shocks & vibration.
consume less power	consume more power

No:.......... Date:....../....../......

SSD

Advantages	Disadvantages.
→ speed – faster and high performance.	→ typically more expensive.
→ more resistant to physical shock or damages.	→ have limited number of write cycles
→ Consume less power, leading to longer battery.	→ data recovery can be more challenging
→ No noise during operation.	

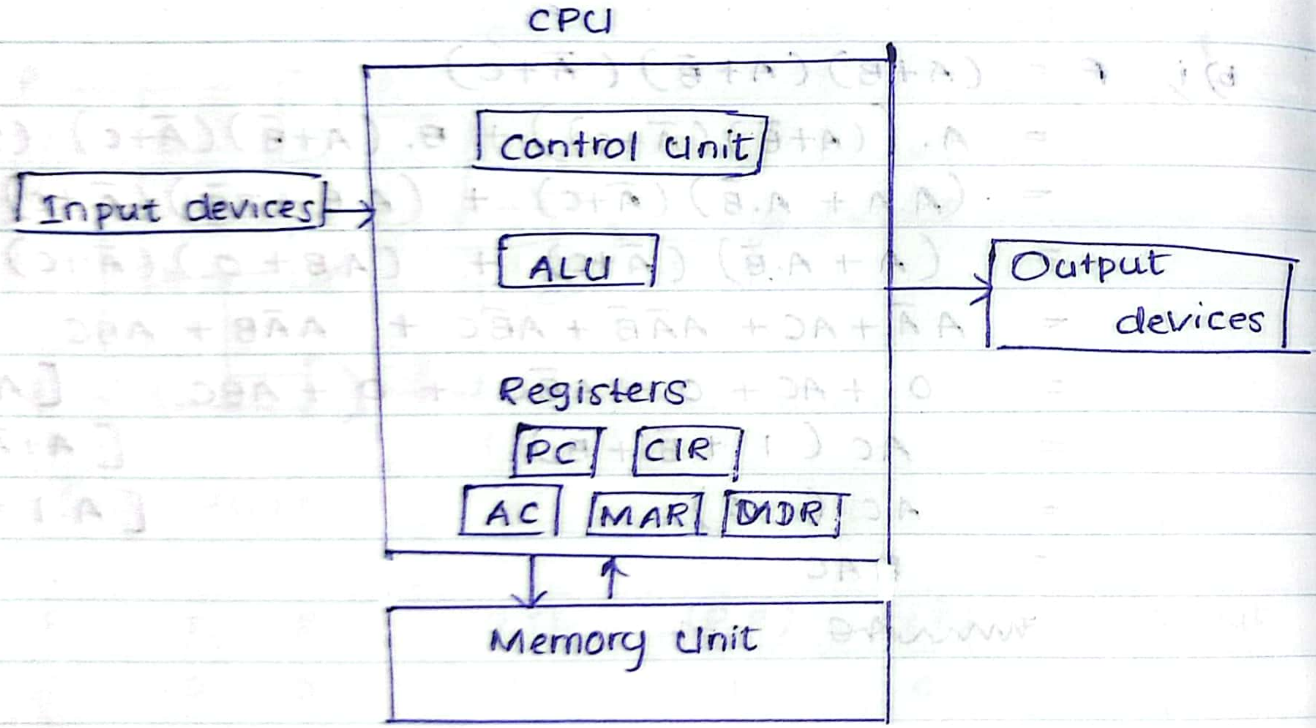
b) i, Register indirect addressing

Using a register to find where data is stored in memory instead of directly saying where to look.

ii) Immediate addressing mode.

The operand is directly specified in the instruction itself. You directly tell the CPU what the data is.

ex :- ADD 5 , where 5 is the data



Von- Neuman Architecture.

- based on the stored-program computer concept
- instruction data, program data are stored in the same memory.
- uses a single processor
- use one memory for both instruction and data.
- following the fetch - decode - execute cycle

fetch - get the instructions from the main memory

decode - figure out what it means.

execute - do whatever the instruction say.

a) i) The data that'll placed MAR will be the address from the PC.

∴ 10010

ii) The data placed in MDR will be the content of the memory address 10010,

∴ 11110001

e) Instruction data, program data are stored in the same memory. It can be define as stored memory concept.

2017/2018

No:.................. Date:......../......./........

2017 – 2018

Q1) a) i) $BC79DE_{16}$

$1011\ 1100\ 0111\ 1001\ 1101\ 1110_2$

57074736_8

B C 7 9 D E

$= (11\times16^5) + (12\times16^4) + (7\times16^3) + (9\times16^2) + (13\times16^1) + (14\times1)$

$= (11\times1048576) + (12\times65536) + (7\times4096) + (9\times256) + (13\times16) +$
(14×1)

$= 11534334_{10}$

ii) $16CDA_{16}$

$0001\ 0110\ 1100\ 1101\ 1010_2$

266332_8

$16CDA_{16}$

$= (1\times16^4) + (6\times16^3) + (12\times\cancel{256}\,16^2) + (13\times16^1) + (10\times1)$

$= (1\times65536) + (6\times4096) + (12\times256) + (13\times16) + (10\times1)$

$= 65536 + 24576 + 3072 + 208 + 10$

$= 93502_{10}$

b) i) $-85 + 11$

85/ 01010101
10101010
+1
10101011
+ 1011
10110110
01001001
+1
01001010_2 = −74

ii) $-72 + 15$

72/ 01001000
10110111
+1
10111000
1111
11000111
00111000
+1
00111001_2

$= -67_{10}$

No: Date: / /

iii) $-100-17$.

100 01100100

10011011

$+1$

10011100

$-$ 10001

10001011

01110100

$+1$

$1110101_2 = -117_{10}$

c) i) $357 = 001101010111_2$

ii) $6579 = 1000010101111001_2$

d) character range

ASCII uses 7 bit range to encode just 128 dintinct characters. Unicode covers more 140000 characters.

space efficiency

ASCII - each character represented by 7 bit.

unicode - uses 2 bytes for each character

language support.

ASCII primarily supports the english language and doesn't cover other language.

e) i) $0.1011_2 = 0 + 0.5 + 0.1875$

$= 0.6875_{10}$

ii) $1101.0111_2 = 13 + 0.25 + 0.125 + 0.0625$

$= 13.4375_{10}$

Q2)

a)

	D	C	B	A	divi 3	divi 8	divi 13	divi 14	F
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	0
2	0	0	1	0	0	0	0	0	0
3	0	0	1	1	1	0	0	0	1
4	0	1	0	0	0	0	0	0	0
5	0	1	0	1	0	0	0	0	0
6	0	1	1	0	1	0	0	0	1
7	0	1	1	1	0	0	0	0	0
8	1	0	0	0	0	1	0	0	1
9	1	0	0	1	1	0	0	0	1
	1	0	1	0	0	0	0	0	0
	1	0	1	1	0	0	0	0	0
	1	1	0	0	1	0	0	0	1
	1	1	0	1	0	0	1	0	1
	1	1	1	0	0	0	0	1	1
	1	1	1	1	1	0	0	0	1

b) $F = \bar{D}\bar{C}BA + \bar{D}CB\bar{A} + D\bar{C}\bar{B}\bar{A} + \cancel{DBCA} + D\bar{C}\bar{B}A + DC\bar{B}\bar{A} + DC\bar{B}A + DCB\bar{A} + DCBA$

c)

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	0 (0)	1 (1)	1 (3)	0 (2)
$\bar{A}B$ 01	0 (4)	0 (5)	1 (7)	1 (6)
AB 11	1 (12)	0 (13)	1 (15)	0 (14)
$A\bar{B}$ 10	0 (8)	1 (9)	1 (11)	0 (10)

$CD + \bar{A}\bar{B}D + \bar{A}BC + A\bar{B}D + AB\bar{C}\bar{D}$

Q3) a) Improving architecture.

Enhancements to processor architecture optimize the way instructions are executed, improving efficiency and speed.

Adding more cores.

Instead of one core faster, manufacturers are putting multiple cores on a single chip. This lets the processor handle many tasks at once, which is great for multitasking.

Improving cashe memory efficiency

Cashe memory stores frequently used data and instructions close to the processor, so it doesn't have to wait for slower main memory. Manufacturers are making this storage area bigger and smarter to speed up data access.

Utilizing Advanced Manufacturing Techniques.

Manufacturers are using better techniques to make processors smaller and more efficient. (fit more transistor on a chip, etc...)

cashe memory.

b) It's a high speed memory sits between cpu and RAM. It'll temperorily store frequently accessed data and instructions. It'll reduce the time. It act as a buffer between CPU and RAM. and reduce the total execution time. Of the program.

c) The hit ratio is 90%. therefore, for 0.9 of accesses are to cashe and 0.1 of accesses are to DRAM.

The average access time is cashe access + DRAM access

$= 50\ ns \times 0.1 + 5ns \times 0.9$

$= 5 + 4.5$

$= 9.5\ ns$

The average access time without cashe is

$= 1.0 \times 50ns$ (Main memory access time)

$= 50\ ns$

The speedup ratio is access time without cashe divide by access time with cashe

$= 50ns / 9.5\ ns$

$= 5.26.$

d) Program counter

It store the memory address of the next instruction to be executed. As the CPU executes each instruction, the PC automatically moves to the next instruction's memory address. This ensures that instructions are executed in the correct sequence and allows for branching to different parts of the program when needed. The PC guides the CPU through the program's execution process.

e) i/ Virtual Memory.

It is a method of using computer harddrive to provide extra memory for the computer. When the computer runs out of physical memory (RAM) It uses some ~~spaces~~ space on the hard drive to store less frequency used data temporarily.

iii/ the interrupt

The interrupts are signals that generated by Hardware or software and need immediate attention. It''ll tell the CPU to stop what it's doing and pay attention to something else. When an interrupt happens, the CPU pause its current task, No.... Date.... / ... /

deals with the interrupt, and then goes back to what it was doing. This helps handle things like input / output operations, time events, or errors withou wasting CPU time constantly checking for them.

4) i, ROM.

ROM is used for storing programs that are permanently resident in the computer. and for tables of constants that do not change in value once the production of the computer is completed.

ii, Instruction set architecture.

It refers to the set of commands or instructions that a processor understands and can execute. It defines the operations that a processor can perform, the formats of instructions ~~to be~~, and the way in which instructions are executed.

iii, Pipelining

It is a technique used in computer processors to overlap the execution of multiple instructions. It breaks down the execution of instruction into several stages, allowing different stages of different instructions to be executed concurrently. This improves overall throughput and performance of the processor.

iv, Parallel Processing.

It involves the simultaneous execution of multiple tasks or instructions by multiple processing units. This can be achieved using multiple processors or cores within a single processor. This enable faster

No: .... Date: ..../......./.......

execution by dividing tasks into smaller parts and processing them concurrently

v/ L1 cashe.

small but fast cashe, located with the CPU core itself. It is used to store frequently accessed instructions and data for quick access by the processer.

b)

Instruction Fetch → Obtain instruction from program storage

↓

Instruction decode → Determine required actions and instruction size.

↓

Operand fetch → Locate and obtain operand data

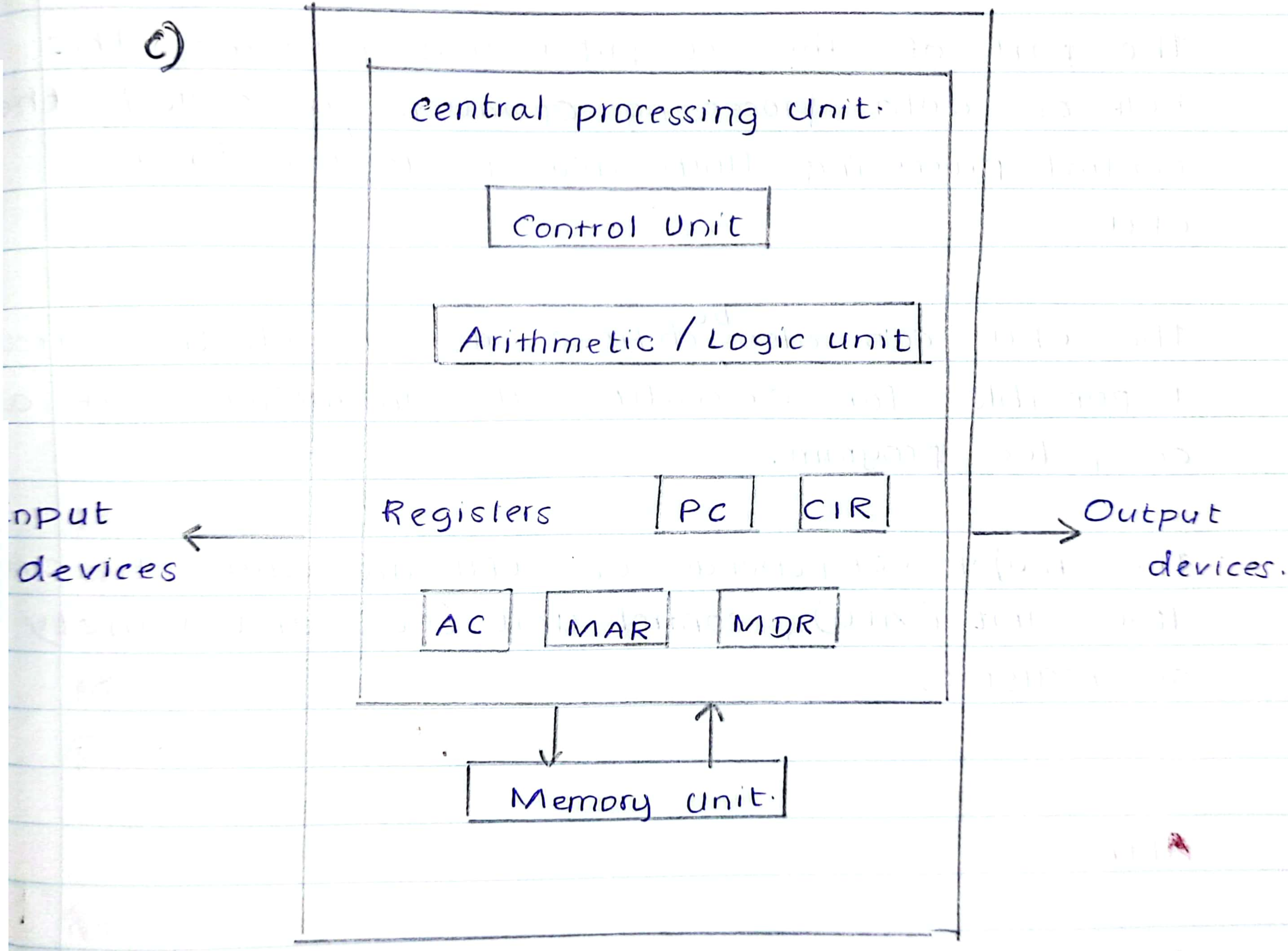
↓

Execute → Compute result value and status

↓

Writeback results. → Deposit results in storage for later use

c)


Central processing unit.
Control Unit
Arithmetic / Logic unit
Registers
PC
CIR
AC
MAR
MDR
Memory unit.
nput devices
Output devices.

d) Instruction data, program data are stored in the same memory. It can be define as stored memory concept.

CISC	RISC
Memory to memory operations	Register to Register operations.
Emphasis on hardware	Emphasis on software.
small code sizes and high cycle per second	Large code size and lower cycles per second.

2016 / 2017

~~20~~ 2016 – 2017

Q1) 1) a) i) $FD7316_{16}$

$1111\ 11010111\ 0011\ 0001\ 0110_2$

ii) 75.5316_{16}

$01110101.0101001100010110_2$

b) 34.625_{10}

2	34
2	17 − 0
2	8 − 1
2	4 − 0
2	2 − 0
	1 − 0

$0.625 \times 2 = 1.250 \rightarrow 1$

$0.250 \times 2 = 0.500 \rightarrow 0$

$0.500 \times 2 = 1.000 \rightarrow 1$

Binary 100010.101_2

Octal. 42.5_8

Hexadecimal $34.625 = 22.A_{16}$

2) a) Sign – Magnitute Representation

One's complement representation

Two's complement representation

b) ~~A−B~~ 54/ 00110110_2

77/ 01001101_2

10110010

+1

10110011_2

A−B

54 − (−77)

00110110

+ 10110011

− ~~1000000011~~

11101001_2

Overflow,

result exceeded the range.

c)

3) a. False b. True c. False d. True e. False

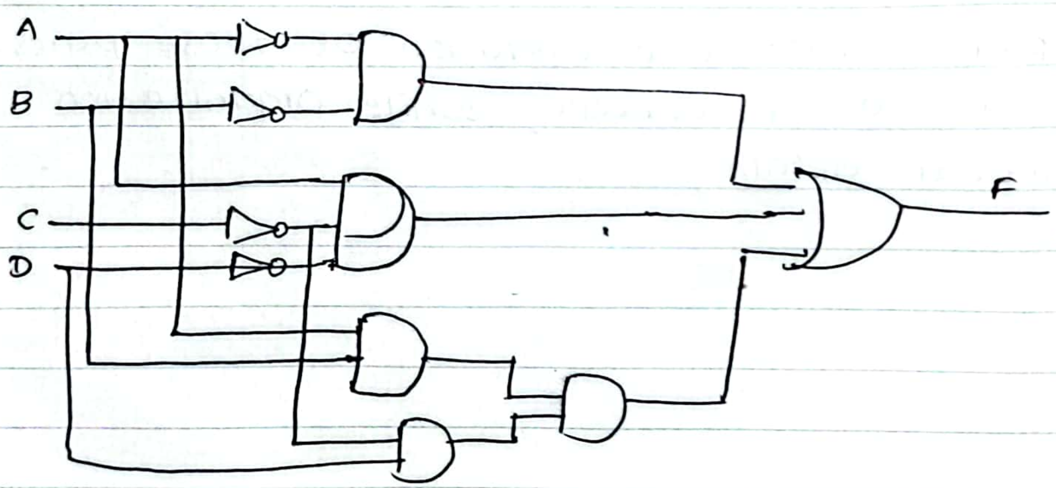
Q8) 1)

CD \ AB	00	01	11	10
00	1		1	1
01			1	
11				
10	1			

F=

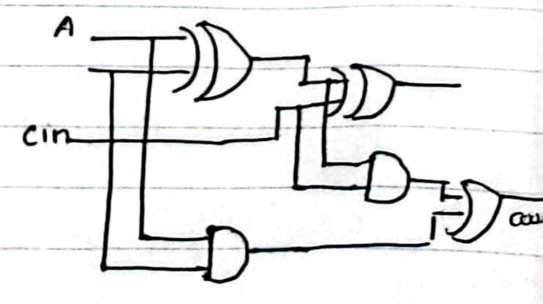
$\bar{A}.\bar{B} + A\bar{C}\bar{D} + AB\bar{C}D.$

logic circuit



2)

cin	a	b	sum	cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Q3) a)

ALU

ALU performs required micro-operations for executing the instructions. In simple words, ALU allows arithmetic (add, subtract, etc...) and logic operations (AND, OR, NOT, etc...) to be carried out.

Control unit.

Control unit of the computer system controls the operations of components like ALU, Memory and Input, Output devices.

* The control unit consists of a program counter (PC) that contains the address of the instruction to be fetched and instruction register (IR) into which instruction are fetched from memory for execution.

No:

Date: / /

Memory Unit.

A Memory unit is a collection of storage cells together with associated circuits needed to transfer information in and out of the storage.

The memory stores binary information in group of bits calls "words"

The internal structure of a memory unit is specified by the number of words it contains and the nu. of bits in each words.

RAM (Random Access Memory)
ROM (Read Only Memory).

Registers

Registers refer~~s~~ to high speed storage areas in the CPU.

the data processed by the CPU are fetched from the registers

Q3) i)

b) Input devices
↓
memory
↓
CPU
↙ ↘
CU ALU

Input devices : Allow user to input data

Control unit : Manages instructions and data flow

ALU : perform arithmatic and logical oper.

Register & cashe : Temperorily store data ~~and instructions currently being used:~~ for fast access

Main Memory : Holds data and instructions currently being used.

Output devices : Display processed information

a) Computer architecture is like the blueprint or design of a computer system, focusing on its structure and how different parts work together logically. Computer Organization deals with the actual physical implementation and operation of those parts. Architecture is about design concept, while organization is about practical details.

SISD : single Instruction single data.
A single data stream is being processed by one instruction stream.
ex: Older generation Computers, mini compu--

SIMD : single instruction Multiple data
Each instruction is executed on a different set of data by different processors under the supervision of a common control unit.

MISD : Multiple instruction single data.
each processor executes a different sequence of instructions.
Non existent.

MIMD : Multiple instructions multiple data.
It have multiple processing units, each with its own instruction stream and data stream.

Instruction set architecture.

It refers to the set of commands or instructions that a processor understands and can execute. It defines the operations that a processor can perform, the formats of instructions ~~to be~~, and the way in which instructions are executed.

Immediate addressing mode.

The operand is directly specified in the instruction itself. You directly tell the CPU what the data is.

ex :- ADD 5 , where 5 is the data

Direct addressing mode.

You tell the CPU where to find the data.

ex :- LOAD from memory address 200. The address is directly specified in the instruction.

indirect addressing mode.

You tell the CPU where to find the address of the data.

Ex :- LOAD from the memory address stored at memory address 300

Pipelining

It is a technique used in computer processors to overlap the execution of multiple instructions. It breaks down the execution of instruction into several stages, allowing different stages of different instructions to be executed concurrently. This improves overall throughput and performance of the processor.

CISC	RISC
Memory to memory operations	Register to Register operations.
Emphasis on hardware	Emphasis on software.
small code sizes and high cycle per second	Large code size and lowest cycles per second.

Shared memory

- multiple processors /processing units share access to a single, global memory space.

- All processors can read from and write to the same memory locations, allowing for easy sharing of data.

No :

- Communication between processors occurs implicity through memory access, and synchroniqation is typically achieved using locks or other mechanisms.

- Ex:- Multi core processors, Where multiple CPU cores share access to the same physical memory.

distributed memory.

- Each processing unit has its own seperate memory space

- Processors communicate by explicity sending messages to each other over a network, as they can't directly access each other's memory.

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- Communication is managed through message passing libraries or protocols like MPI (Message Passing Interface)

- Ex:- Cluster computing system where each node is in the cluster has its own memory and communicates with other nodes via a network.